ENVIRONMENTAL PRODUCT DECLARATION

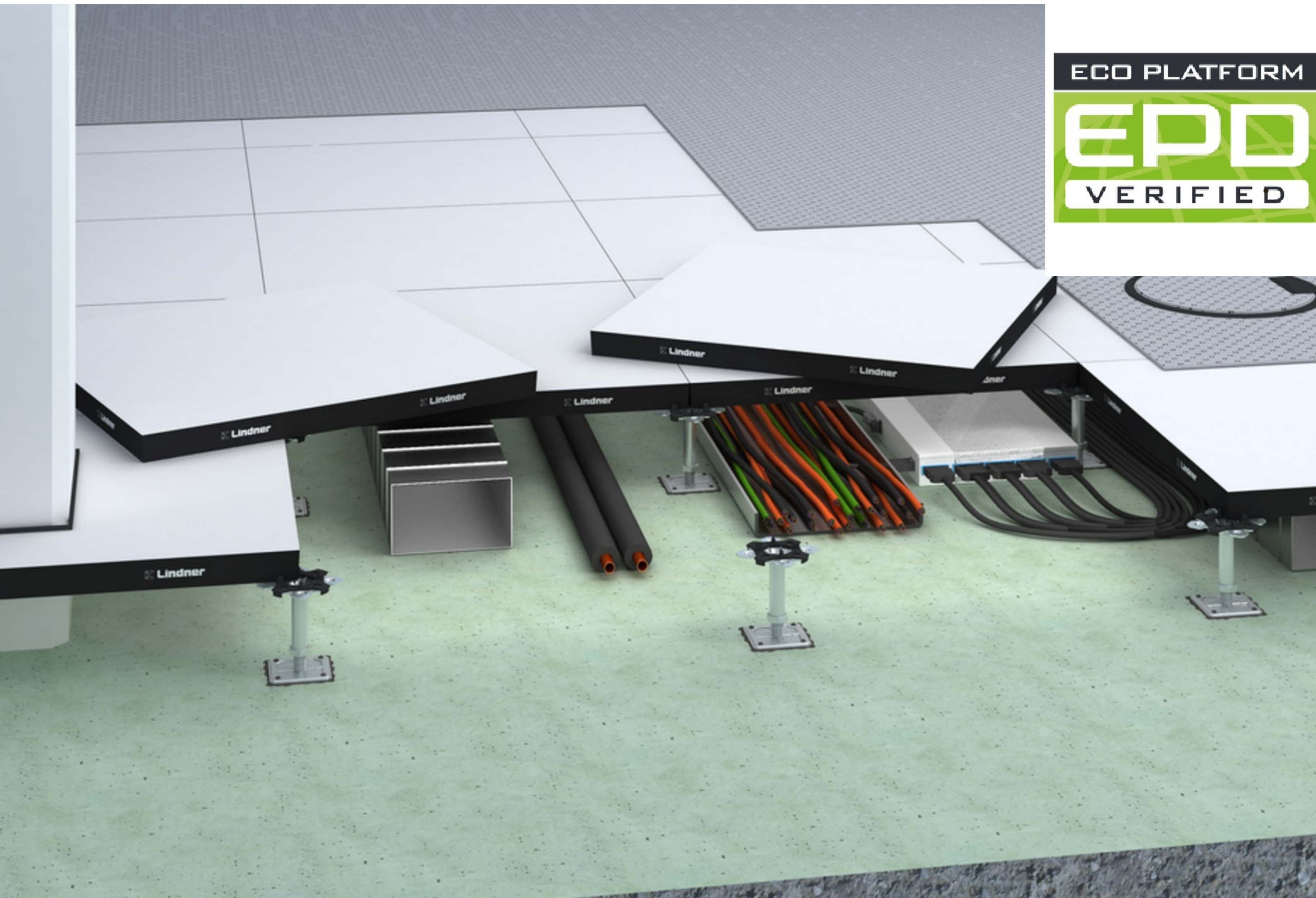
as per *ISO 14025* and *EN 15804+A2*

Owner of the Declaration	Lindner Group
Publisher	Institut Bauen und Umwelt e.V. (IBU)
Programme holder	Institut Bauen und Umwelt e.V. (IBU)
Declaration number	EPD-LIN-20260123-IBA2-EN
Issue date	12.06.2026
Valid to	11.06.2031

Raised Floor Panel, Type NORTEC G 38 ST

Lindner Group

www.ibu-epd.com | https://epd-online.com



1. General Information

Lindner Group	Raised Floor Panel, Type NORTEC G 38 ST
Programme holder IBU – Institut Bauen und Umwelt e.V. Hegelplatz 1 10117 Berlin Germany	**Owner of the declaration** Lindner Group Bahnhofstraße 29 94424 Arnstorf Germany
Declaration number EPD-LIN-20260123-IBA2-EN	**Declared product / declared unit** 1 m² of the NORTEC G38 ST raised floor system.
This declaration is based on the product category rules: System floors, 01.08.2021 (PCR checked and approved by the SVR) **Issue date** 12.06.2026 **Valid to** 11.06.2031	**Scope:** This EPD pertains to the NORTEC G38 ST raised floor panel, consisted of gypsum panels with an approximate density of 1,399 kg/m³ and a thickness of 38.4 mm. The raised floor panels are manufactured at the Lindner plant in Dettelbach, and the production data are collected for the year 2024. The owner of the declaration shall be liable for the underlying information and evidence; the IBU shall not be liable with respect to manufacturer information, life cycle assessment data and evidence. The EPD was created according to the specifications of EN 15804+A2. In the following, the standard is abbreviated as *EN 15804*.

Verification

The standard EN 15804 serves as the core PCR
Independent verification of the declaration and data according to ISO 14025:2011
☐ internal ☒ external

Dipl.-Ing. Hans Peters
(Chairman of Institut Bauen und Umwelt e.V.)

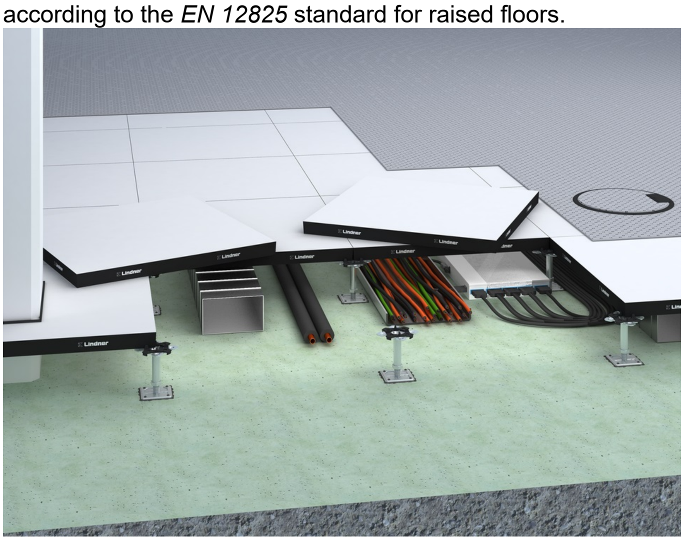
Dr. Martina Bender
(Managing Director Institut Bauen und Umwelt e.V.)

Erik Poppe,
(Independent verifier)

2. Product

2.1 Product description/Product definition

The declared product, Lindner raised floor panel, type NORTEC G38 ST, is a fibre-reinforced calcium sulphate panel,serving as a key component of the dry raised floor system ,NORTEC. This product features a thickness of 38.4 mm and a bulk density of approximately 1,399 kg/m³. The raised floor panels are typically manufactured instandard dimensions of 600 x 600 mm. The NORTEC panels are assembled using a special serrated groove along the panel edges, forming a closed base layer. The raised floor panel, type NORTEC, was developed and tested according to the *EN 12825* standard for raised floors.



The use of this product is subject to the respective national regulations where it is installed, such as the building codes of the federal states in Germany and the technical provisions derived from this legislation.
The substructure (pedestals and profiles) and surface covering are not included in the life cycle assessment (LCA) and Environmental Product Declaration (EPD); they are evaluated through separate LCA values and EPDs.

For the use and application of the product the respective national provisions at the place of use apply, in Germany for example the building codes of the federal states and the corresponding national specifications.

2.2 Application

The NORTEC G38 ST raised floor panels, made of fibre reinforced calcium sulphate, are designed primarily for interior use in public, commercial, and private buildings. They create hollow installation spaces beneath the base layer, which allows for flexible use and the accommodation of various installations, supply lines, and disposal systems. Access to these cavities can be provided through inspection openings or raised floor routes. When combined with additional components, these panels form the dry raised floor system,type NORTEC G38 ST, which can be covered with all standard floor coverings. However, it is essential to coordinate the different system variants to ensure compatibility.

2.3 Technical data

Technical data

The technical specifications of the NORTEC G38 ST within the scope of the EPD are as follows:

Name	Value	Unit
Layer thickness base courese (from - to)	0.0384	m
Grammage / panel weight	53.72	kg/m^2
Density of the base course	1399	kg/m^3
Point load Statics (EN 12825)*	2 - 5	kN
Fire protection * (DIN EN 13501-01:2010-01, classification of building products and types of construction with regard to their fire behaviour)	A1,A2	-
Fire protection * (DIN 4102-1:1998: Fire behaviour of building materials and components)	F30/F60 REI30 REI60	-
Sound insulation insulation (laboratory values; VDI 3762 is to be observed)* standard side noise level difference D nfw	48 - 57	dB
Sound insulation (laboratory values; VDI 3762 is to be observed)* sound insulation Rw	62	dB
Sound insulation (laboratory values; VDI 3762 is to be observed)* standard side noise level L nfw	73 - 47	dB
Sound insulation (laboratory values; VDI 3762 is to be observed)* footfall noise level reduction ΔL w	12 - 37	dB

* The listed values show the complete testing range of the NORTEC G38 ST raised floor panel. Values for the specific raised floor system were evidenced by individual test reports

Product performance characteristics are in accordance with the relevant technical specifications (no CE approval mark).

2.4 Delivery status

NORTEC G38 ST raised floor panels are delivered stacked on a pallet. The stack height depends on the thickness of the panels and the type of covering applied.

2.5 Base materials/auxiliary materials

Name	Value	Unit
Gips	88.54	%
Cellulose fibers	4.08	%
Auxiliaries	1.63	%
Steel	5.75	%

For greater transparency, the content declaration of the product and base materials for the product are as follows:
A NORTEC G38 ST raised floor panel consists of the following base materials:

- REA-Gypsum (α-hemihydrate)
- Calcined Gypsum (β-hemihydrate)
- Cellulose fibers
- Water

These materials are briefly described in more detail below
REA-Gypsum (α-hemihydrate)
Gypsum ($CaSO_4$ * $2H_2O$) is a sulfate mineral in a powder form. The REA-gypsum used at the
production site is produced industrially by desulfurization of the flue gases when coal is burned.

Calcined Gypsum (β-hemihydrate)
Calcined gypsum ($CaSO_4$* $1/2H_2O$), generated as rest material from production through grinding, it is
rehashed and then inserted into the production flow. Through this recycling process, we reserve

natural gypsum resources.

Cellulose fibres
The required cellulose fibers are produced exclusively from secondary materials, including recycled paper and old papers sourced from the Lindner Company in Arnstorf. After high-consistency pulping, thinning with water, and subsequent impurity removal, the fiber is then brought to the desired density. As a result, most NORTEC variants are certified FSC™ Recycled 100%.

Water
The majority of the process water is sourced from well water pumped on-site, with only a small amount of district water used. After production, the water is treated and recycled back into the process as circulating process water

It is important to note that none of the mentioned materials are substances from the "Candidate List of Substances of VeryHigh Concern for Authorisation" (SVHC) with a masspercentage greater than 0.1% of the construction product'smass. Additionally, the construction product is not classified asa substance or mixture under the chemical law (*REACH*).Therefore, it can be stated that:

- "This product/article/at least one component containssubstances listed in the candidate list (date: 04.02.2026 according to https://echa.europa.eu/de/candidate-listtable) exceeding 0.1% by mass: NO."
- "This product/article/at least one component containsother carcinogenic, mutagenic, reprotoxic (CMR)substances in categories 1A or 1B that are not on thecandidate list, exceeding 0.1% by mass: NO."
- "Biocide products were added to this constructionproduct, or it has been treated with biocide products(this then concerns a treated product as defined by the(EU) *Regulation on Biocidal Products No. 528/2012*): NO."

2.6 Manufacture
Production and treatment of the raised floor panel involves mixing essential raw materials such as Alpha Hemihydrate, Beta Gypsum, and cellulose fibres. After adding water, these materials are pressed into stable panels under high pressure.The panels are then dried and cut or milled to the required formats. An optional steel is applied at the bottom of the panel
In further production steps, a serration is milled into the sides of the panels.
Lindner Group operates a quality management system in accordance with */EN ISO 9001/.*

2.7 Environment and health during manufacturing
The production of fibre-reinforced calcium sulfate panels takes place in environmentally approved facilities. The process water used is recirculated as much as possible within a closed-loop system. The gypsum waste generated is largely reintroduced into the material cycle within the facility. Lindner Group operates an energy management system in accordance with */EN ISO 50001/* and an environmental management system in accordance with */EN ISO 14001/.*

2.8 Product processing/Installation
The raised floor panels of type NORTEC G38 ST delivered to the construction site are combined with additional individual components to create a complete flooring system. For further instructions, please refer to the hollow floor installation guidelines.
Installation must be carried out by trained personnel.

2.9 Packaging
Raised floor panels are delivered stacked on pallets, specifically on single pallets with two runners for the FINAL packaging.They are wrapped in cardboard, strapped with plastic tape,and, if necessary, covered with plastic film to protect them from moisture during transport. Each pallet can reach a maximum height of 4,000 mm and can be stacked up to four pallets high.Additionally, supports and other individual components are stacked or layered in cardboard packaging, ensuring that all items remain secure and protected throughout the delivery process.
The packaging material is easily separable and can, where applicable, be reused or repurposed. The remaining material,once correctly separated, should be collected and taken to the regional recycling centre. Residual materials must be disposed of in accordance with relevant national regulations. The packaging specifications for all standard Lindner products are detailed in the packaging data sheets.

2.10 Condition of use
When used as designated, no material changes in composition are to be anticipated during the use phase.
The service life of floor panels is planned to match the entire lifespan of the building. Based on long-term experience, no significant changes in material composition are expected during the use of the NORTEC G38 ST raised flooring panel.

2.11 Environment and health during use
No health hazards or impairments are expected based on current knowledge when the NORTEC raised floor panel are used as intended and in accordance with their designated application. For further details, refer to Section 7 (Indoor Air Comfort Gold Label).
According to the current state of knowledge, proper use of the described products does not result in harmful exposure to air, water, or soil.

2.12 Reference service life
A reference service life according to *ISO 15686* cannot be calculated for this product. Therefore, the technical service life is derived from the table "Service Life of Components for Life Cycle Analysis According to the Rating System for Sustainable Construction (Bewertungs system Nachhaltiges Bauen – BNB) –Code No. 352.911" by the Federal Office for Construction and Regional Planning *(BBSR)*. The BNB assumes that raised floor systems and panels will have a service life of over 50 years.This stated service life is contingent upon proper use,maintenance, and care.
The document also describes the influences on the aging of the product when applied according to established technical guidelines.
Description of the influences on the ageing of the product when applied in accordance with the rules of technology.

2.13 Extraordinary effects
Fire

Fire protection
Fibre-reinforced calcium sulphate panels, used as raised floor panels, are classified as "non-flammable" according to */EN13501-1/* and fall into building material class A1.
Depending on the board thickness, the declared raised floor system is classified in classes REI 30, REI 60 and F 30, F 60 in accordance with *EN13501-2* and *DIN 4102-2.*

Name	Value	Unit
Building material class	A1	-
Smoke gas development	S1	-
Burning droplets	d0	-

Water
Lindner's NORTEC raised floor panels are intended for indoor use and should generally not come into contact with water. Short-term exposure to moisture will not damage the system, provided that the panels are allowed to dry completely afterwards. Prolonged exposure to large amounts of water will not cause the leaching of substances that could pollute water courses.
However, such exposure may impair the technical properties of the panels, as Lindner raised floor panels are not water resistant. In very damp or wet conditions, the panels may swell, and the supports may corrode.

Mechanical destruction
The durability and functionality of the panel system will be impaired in the event of mechanical damage. Depending on the extent of the damage, affected areas can be repaired or replaced with new panels without impairing overall functionality

2.14 Re-use phase
Direct reuse of the panels immediately after removal is not feasible due to the technical characteristics of the panel edges. Therefore, two end-of-life (EoL) scenarios have been considered for the panels with a steel sheet:
• Reuse: Panels are reused as new floor panels with smaller dimensions.
• Landfilling: Panels are not used again and are sent to landfill.
The reuse scenario reflects a specific take-back and recovery route implemented by Lindner, in which dismantled panels are treated and reused as smaller panels. If this route is not available, common practice depends on local waste management infrastructure and the condition of the panels after use. Typical options include selective dismantling for possible reuse, sorting and separation of material fractions in construction and demolition waste treatment facilities, recycling of steel components where feasible, and disposal of the gypsum-based fraction. Where no suitable recycling or recovery route exists, landfilling is a common option.
Accordingly, the scenarios modelled in the EPD represent both a recovery route with reuse potential and a conservative end of-life route reflecting common market practice.

2.15 Disposal
Any remnants of raised floor panels from construction or dismantling activities that cannot be reused or recycled should be disposed of in accordance with the applicable waste codes according to the European Waste Catalogue *(EWC)* (Commission Decision 2000/532/EC as amended by Commission Decision 2014/955/EU). Depending on the material composition of the panels, the following waste codes apply: 17 08 02: gypsum-based construction materials (other than those classified under 17 08 01)

2.16 Further information
For additional product information, please visit: www.LindnerGroup.com.

3. LCA: Calculation rules

3.1 Declared unit
The declared unit refers to 1 m² raised floor panel, type NORTEC G38 ST, The raised floor system has a panel thickness of
38.4 mm and an average density of 1,399 kg/m³.
Supports and surface coverings are not included in the life cycle assessment due to their varied types and specifications.Possible surface coverings include parquet, linoleum, carpet,etc.

Declared unit and mass reference

Name	Value	Unit
Declared unit	1	m^2
Grammage	53.72	kg/m^2
Gross density	1,399	kg/m^3
Layer thickness	0.0384	m

3.2 System boundary
The life cycle analysis for the NORTEC G38 ST raised floor panel covers "cradle to grave with options" (A1–A3, A4 , A5 + C1-C4+ D) Specifically, the following processes were included in information module A1–A3 for the production of the NORTEC raised floor system:

- Raw material provision (A1)
- Transport of raw, auxiliary materials, and consumables to the plant (A2)
- Production processes at the plant, including energy costs (electricity, thermal energy), water consumption,production of packaging materials and disposal of residual substances (A3)
- Transport from the factory gate to the construction site (A4)
- Disposal of packaging materials (A5)
- Dismantling of the panels from the building (C1)
- Transportation of discarded panels to facilities for processing, recycling, or disposal (C2)
- Processing waste materials from the deconstruction stage, including preparation for recycling and energy recovery where applicable (C3)
- Disposal of waste materials that cannot be reused or recycled, including physical pre-treatment and management of the disposal site for landfilling scenarios(C4)
- Reporting on the benefits and loads associated with the reuse, recovery, and recycling of materials beyond the system boundary, including energy recovery from the disposal of materials (D)

3.3 Estimates and assumptions
In the EPD, the GWP for the electricity mix in modules A1–A3 is 0.67 CO_2e/kWh, while the thermal energy mix accounts for0.223 kg CO_2e/kWh. With the exception of Modules A1 to A3, which describe the manufacturing process and are therefore based on actual foreground data, all other modules are calculated using assumptions, referred to as scenarios. As mentioned before, the developed End-of-Life scenarios are as follows:

Scenario 0 – Reuse:
Panels are carefully dismantled and sorted on-site. A loss of material occurs during the reuse process after milling and grinding the edges. Assumptions and calculations regarding the grinding and milling of plaster boards are documented in the project background report. When the edges of a gypsum panel are milled, their area is reduced, which in turn results in a reduction in weight, forming the required input for LCA modelling. The weight loss can be calculated using the area removed during milling and the grammage (weight per unit area) of the gypsum panel, as documented in the project background report.

Scenario 1 – Landfilling:
Gypsum panels are dismantled and sent directly to landfill without recycling, remaining as waste and not entering any

recovery process.

It is also assumed that all waste incineration plants have an R1 factor greater than 0.6, indicating a high level of efficiency in energy recovery from waste. This assumption is based on statistical data from waste incineration plants in Northern Europe, as detailed in the CEWEP Energy Report II (Status 2007–2010).
Apart from the End-of-Life scenarios, for all transportation-related modules including A2 for raw material extraction, A4 for transport from the production site to the construction site, and C2 for waste processing the 'GLO: LKW-Zug/Sattel-Zug, Euro 6 A-C' process is used, assuming a scalable distance of 100 km. This reflects the reality that a significant portion of raw materials and NORTEC system products are delivered within Germany, covering a substantial delivery radius from Dettelbach, near Würzburg.

3.4 Cut-off criteria

Some materials were excluded from the inventory and analysis both due to the small and negligible amounts and also due to the gaps in the *Sphera MLC* (fka GaBi) *CUP 2023.02* databases. These exclusions were considered insignificant to the overall environmental performance of the product and do not affect the results of the assessment. It is estimated that these omitted processes would contribute less than 5% to the impact categories considered.

3.5 Background data

The software system for holistic balancing, LCA for Experts Version 10.7.1.28, developed by *Sphera*, was used to model the life cycle of the product in question *(Sphera MLC* (fka GaBi) *CUP 2023.02)*. Data required for the upstream chain, where specific details were unavailable, were sourced from the LCA for Experts database.

3.6 Data quality

The background data gives rise to some minor uncertainties owing to the provision of LCA FE databases and which must be taken into consideration when interpreting the results. The data quality assessment considers technical, time-based, and geographical representativeness. The foreground data is the production data based on 2024. The background data is based on the LCA data set. The data quality can be regarded as good.

3.7 Period under review

Production data for this assessment were collected from the Lindner plant in Dettelbach, Germany, for the year 2024.

3.8 Geographic representativeness

Country or region in which the declared product system is manufactured, used or handled at the end of the product's life span: Europe

3.9 Allocation

The Lindner Group's gypsum-based production line manufactures various products, including the declared NORTEC G38 ST raised floor panel. Data collection for thermal and electrical energy use, as well as raw material consumption, focuses specifically on the declared product and unit. Since the production processes for NORTEC fibre-reinforced plasterboards, are not segmented into distinct sub processes,all products produced on the same line undergo identical production steps. The only differences between the NORTEC G38 ST raised floor panels and other products are related to the panel's density and thickness.To accurately allocate resources such as raw materials and energy used in production, these factors are assigned proportionally based on a virtual product that represents an average of all variants, considering sales figures and production amounts through a weighted average approach.
Each of these allocations must reference the modules in which they are performed. Additionally, when considering multiple products processed together, a physical classification method is utilized to allocate environmental impacts based on the consumption of raw materials, packaging, energy, and water relative to total production output.The methodology comprehensively incorporates end-of-life considerations. Specific allocation methods are clearly documented in the project report, ensuring that the reported data accurately reflects the resources consumed for the production of the NORTEC panels while maintaining transparency and consistency throughout the evaluation process.In detail, allocations relevant for calculation must encompass several key areas:

- Allocation of energy, auxiliary, and operating materials used for individual products in a factory.
- Allocation of co-production processes.
- Allocation in the use of recycled and/or secondary raw materials.
- Loads and benefits beyond the system boundary from recycling or energy recovery of packaging materials and production waste.
- Loads and benefits beyond the system boundary from recycling or energy recovery from the end of life of the product.

3.10 Comparability

Basically, a comparison or an evaluation of EPD data is only possible if all data sets to be compared were created according to *EN 15804* and the building context and product-specific performance characteristics are taken into account. The */LCA FE database/* was used (see section 8 References).

4. LCA: Scenarios and additional technical information

Characteristic product properties of biogenic carbon

The biogenic carbon content quantifies the amount of biogenic carbon present in NORTEC G38 ST as it leaves the factory gate and must be separately declared for both the product and its accompanying packaging.

Information on describing the biogenic carbon content at factory gate

Name	Value	Unit
Biogenic carbon content in product	0.91	kg C
Biogenic carbon content in accompanying packaging (Wooden palette+oak wood+Cardboard)	0.36	kg C

Note: 1 kg of biogenic carbon is equivalent to 44/12 kg of CO_2.

Scenario 0: 100% Reuse
Scenario 1: 100% Landfill
The raised floor panels are delivered on wooden pallets and protected with cardboard packaging and PE film. During installation on the building site, the packaging is removed, generating approximately 0.76 kg/m² of wooden pallet, 0.03 kg/m² of oak wood, 0.09 kg/m² of cardboard, and 0.006 kg/m² of PE film per declared unit. The packaging waste is assigned to Incineration.

Transport from the gate to the site (A4)

Name	Value	Unit
Fuel type	Diesel Mix	
Litres of fuel	0.155	l/100km
Transport distance	100	km
Capacity utilisation (including empty runs)	60	%
Gross density of products transported	1,399	kg/m^3
Total weight of the transported good (packed)	54.61	kg

A reference service life cannot be calculated for this product. A technical service life of around 50 years is assumed according to the BNB.

Reference service life

Name	Value	Unit
Life Span (according to BBSR)	50	a

End of life (C1-C4)

Scenario 0: 100% Reuse
Scenario 1: 100% Landfill
Scenario 2: 100% Recycling

Name	Value	Unit
Collected mixed waste	53,72	kg
Sc 0 Materials for Reuse	51.06	kg
Sc 0 Waste to Landfill	2.65	kg
Sc 1 Waste to Landfill	53.72	kg

Reuse, recovery and/or recycling potentials (D), relevant scenario information

Name	Value	Unit
Credit from material Sc 0 - gypsum board	51.06	kg
Electrical Energy Recovery	2.05	KJ
Thermal Energy Recovery	3.69	KJ

Scenario 0 – Reuse : The panel is reused as new panels. The credit reflects the avoided production of new raised floor panels.

5. LCA: Results

Information on the environmental impacts is established using the characterization factors according to the EF 3.1 method, as required by *PCR Part A*, section 8.2.1. Long-term emissions are not taken into consideration. The characterization factors applied comply with the requirements of Annex C of *DIN EN 15804*.

Scenario 0: 100% Reuse - represented by C1, C2, C3, C4, D
Scenario 1: 100% Landfill - represented by C1, C2/1, C3/1, C4/1, D/1

In both Scenario 0 (Reuse) and Scenario 1 (Landfilling), the panels are manually demounted without using energy or resources; therefore, C1 is declared but has a value of 0 for both scenarios.

DESCRIPTION OF THE SYSTEM BOUNDARY (X = INCLUDED IN LCA; MND = MODULE OR INDICATOR NOT DECLARED; MNR = MODULE NOT RELEVANT)

Product stage			Construction process stage		Use stage							End of life stage				Benefits and loads beyond the system boundaries
Raw material supply	Transport	Manufacturing	Transport from the gate to the site	Assembly	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
A1	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
X	X	X	X	X	MND	MND	MNR	MNR	MNR	MND	MND	X	X	X	X	X

RESULTS OF THE LCA - ENVIRONMENTAL IMPACT according to EN 15804+A2: 1 m2 NORTEC G38 ST

Parameter	Unit	A1-A3	A4	A5	C1	C2	C2/1	C3	C3/1	C4	C4/1	D	D/1
GWP-total	kg CO_2 eq	1.5E+01	4.71E-01	1.44E+00	0	4.64E-01	4.63E-01	3.42E+00	0	2.17E-01	4.39E+00	-1.9E+01	-4.42E-01
GWP-fossil	kg CO_2 eq	2E+01	1.3E-01	4.72E-02	0	4.49E-01	4.46E-01	1.08E-02	0	3.93E-02	7.95E-01	-1.89E+01	-4.38E-01
GWP-biogenic	kg CO_2 eq	-4.98E+00	3.38E-01	1.4E+00	0	1.02E-02	1.47E-02	3.41E+00	0	1.77E-01	3.59E+00	-2.77E-02	-3.52E-03
GWP-luluc	kg CO_2 eq	6.53E-03	2.83E-03	2.86E-05	0	4.21E-03	2.78E-03	1.17E-06	0	1.24E-04	2.51E-03	-3.43E-03	-4.38E-05
ODP	kg CFC11 eq	2.35E-11	1.17E-13	1.45E-13	0	6.36E-14	1.15E-13	1.98E-13	0	1.01E-13	2.05E-12	-2.62E-11	-6.15E-12
AP	mol H^+ eq	2.81E-02	7.26E-04	2.53E-04	0	7.78E-04	7.14E-04	2.3E-05	0	2.83E-04	5.72E-03	-2.58E-02	-4.71E-04
EP-freshwater	kg P eq	1.32E-05	1.11E-06	5.01E-08	0	1.66E-06	1.1E-06	4.01E-08	0	8.03E-08	1.63E-06	-1.08E-05	-1.36E-06
EP-marine	kg N eq	8.88E-03	2.94E-04	7.6E-05	0	3.06E-04	2.89E-04	5.49E-06	0	7.31E-05	1.48E-03	-8.01E-03	-1.71E-04
EP-terrestrial	mol N eq	9.66E-02	3.36E-03	1.07E-03	0	3.51E-03	3.31E-03	5.74E-05	0	8.04E-04	1.63E-02	-8.7E-02	-1.81E-03
POCP	kg NMVOC eq	2.69E-02	6.55E-04	2.06E-04	0	6.94E-04	6.44E-04	1.47E-05	0	2.21E-04	4.46E-03	-2.42E-02	-4.35E-04
ADPE	kg Sb eq	5.34E-07	3.42E-08	1.62E-09	0	3.09E-08	3.36E-08	1.66E-09	0	1.84E-09	3.73E-08	-4.91E-07	-4.31E-08
ADPF	MJ	2.67E+02	6.42E+00	4.17E-01	0	6.32E+00	6.32E+00	2.26E-01	0	5.31E-01	1.07E+01	-2.52E+02	-6.81E+00
WDP	m^3 world eq deprived	2.47E-01	2.48E-03	1.59E-01	0	5.42E-03	2.44E-03	2.4E-03	0	4.38E-03	8.86E-02	-2.73E-01	-6.59E-03

GWP = Global warming potential; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential of land and water; EP = Eutrophication potential; POCP = Formation potential of tropospheric ozone photochemical oxidants; ADPE = Abiotic depletion potential for non-fossil resources; ADPF = Abiotic depletion potential for fossil resources; WDP = Water (user) deprivation potential)

RESULTS OF THE LCA - INDICATORS TO DESCRIBE RESOURCE USE according to EN 15804+A2: 1 m2 NORTEC G38 ST

Parameter	Unit	A1-A3	A4	A5	C1	C2	C2/1	C3	C3/1	C4	C4/1	D	D/1
PERE	MJ	1.72E+01	4.31E-01	1.29E+01	0	4.58E-01	4.24E-01	1.35E-01	0	8.66E-02	1.75E+00	-1.42E+01	-2.99E+00
PERM	MJ	1.28E+01	0	-1.28E+01	0	0	0	0	0	0	0	0	0
PERT	MJ	3E+01	4.31E-01	9.41E-02	0	4.58E-01	4.24E-01	1.35E-01	0	8.66E-02	1.75E+00	-1.42E+01	-2.99E+00
PENRE	MJ	2.68E+02	6.44E+00	5.37E-01	0	6.35E+00	6.33E+00	2.26E-01	0	5.31E-01	1.08E+01	-2.53E+02	-6.81E+00
PENRM	MJ	1.2E-01	0	-1.2E-01	0	0	0	0	0	0	0	0	0
PENRT	MJ	2.68E+02	6.44E+00	4.17E-01	0	6.35E+00	6.33E+00	2.26E-01	0	5.31E-01	1.08E+01	-2.53E+02	-6.81E+00
SM	kg	2.26E+00	0	0	0	0	0	0	0	0	0	2.53E+01	0
RSF	MJ	0	0	0	0	0	0	0	0	0	0	0	0
NRSF	MJ	0	0	0	0	0	0	0	0	0	0	0	0
FW	m^3	2.22E-02	3.83E-04	3.74E-03	0	4.97E-04	3.77E-04	1.09E-04	0	1.34E-04	2.72E-03	-2.22E-02	-1.06E-03

PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy resources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water

RESULTS OF THE LCA – WASTE CATEGORIES AND OUTPUT FLOWS according to EN 15804+A2: 1 m2 NORTEC G38 ST

Parameter	Unit	A1-A3	A4	A5	C1	C2	C2/1	C3	C3/1	C4	C4/1	D	D/1
HWD	kg	4.61E-08	1.08E-11	8.69E-12	0	1.91E-11	1.07E-11	0	0	1.16E-11	2.34E-10	-2.49E-08	-2.37E-10
NHWD	kg	1.53E-01	9.63E-04	2.93E-02	0	9.66E-04	9.47E-04	1.66E-04	0	2.66E+00	5.38E+01	-1.45E-01	-3.93E-03
RWD	kg	6.31E-03	8.47E-06	2.06E-05	0	1.17E-05	8.33E-06	3.6E-05	0	6.06E-06	1.23E-04	-6.08E-03	-3.09E-04
CRU	kg	0	0	0	0	0	0	5.11E+01	0	0	0	0	0
MFR	kg	1.6E-01	0	0	0	0	0	0	0	0	0	0	0
MER	kg	0	0	0	0	0	0	0	0	0	0	0	0
EEE	MJ	0	0	1.85E+00	0	0	0	0	0	0	0	0	0
EET	MJ	0	0	3.33E+00	0	0	0	0	0	0	0	0	0

HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed; CRU = Components for re-use; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy

RESULTS OF THE LCA – additional impact categories according to EN 15804+A2-optional: 1 m2 NORTEC G38 ST

Parameter	Unit	A1-A3	A4	A5	C1	C2	C2/1	C3	C3/1	C4	C4/1	D	D/1
PM	Disease incidence	6.43E-07	4.68E-09	1.65E-09	0	5.17E-09	4.6E-09	1.93E-10	0	3.48E-09	7.04E-08	-4.47E-07	-3.47E-09
IR	kBq U235 eq	5.35E-01	9.07E-04	3.28E-03	0	1.72E-03	8.93E-04	5.99E-03	0	7E-04	1.42E-02	-5.07E-01	-3.27E-02
ETP-fw	CTUe	4.52E+01	4.69E+00	1.88E-01	0	4.53E+00	4.61E+00	6.29E-02	0	2.87E-01	5.81E+00	-3.92E+01	-1.22E+00
HTP-c	CTUh	1.08E-08	9.35E-11	1.56E-11	0	9.19E-11	9.2E-11	3.33E-12	0	4.46E-11	9.03E-10	-1.02E-08	-8.79E-11
HTP-nc	CTUh	9.51E-08	3.92E-09	8.11E-10	0	4.08E-09	3.86E-09	5.31E-11	0	4.71E-09	9.53E-08	-8.68E-08	-2.18E-09
SQP	SQP	2.6E+02	2.29E+00	1.29E-01	0	2.62E+00	2.25E+00	8.88E-02	0	1.29E-01	2.61E+00	-1.16E+01	-2.07E+00

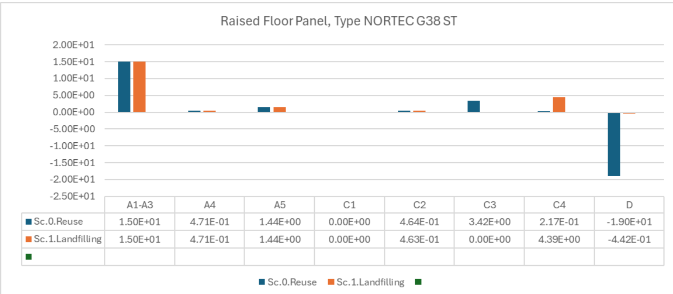
PM = Potential incidence of disease due to PM emissions; IR = Potential human exposure efficiency relative to U235; ETP-fw = Potential comparative toxic unit for ecosystems; HTP-c = Potential comparative toxic unit for humans (carcinogenic); HTP-nc = Potential comparative toxic unit for humans (not carcinogenic); SQP = Potential soil quality index

Disclaimer 1 – for the indicator “Potential human exposure efficiency relative to U235”. This impact category deals mainly with the eventual impact of low-dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure or radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, radon and from some construction materials is also not measured by this indicator.

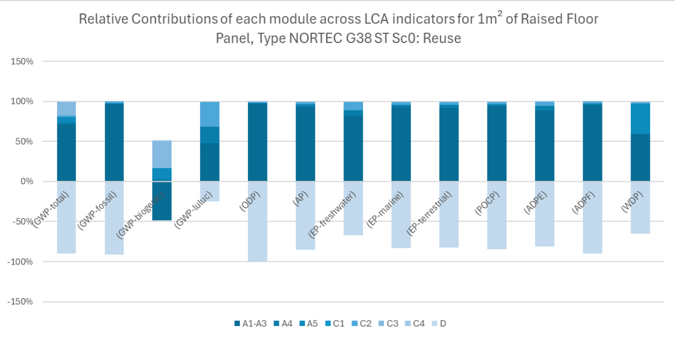
Disclaimer 2 – for the indicators “abiotic depletion potential for non-fossil resources”, “abiotic depletion potential for fossil resources”, “water (user) deprivation potential, deprivation-weighted water consumption”, “potential comparative toxic unit for ecosystems”, “potential comparative toxic unit for humans – carcinogenic”, “Potential comparative toxic unit for humans - not carcinogenic”, “potential soil quality index”. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high as there is limited experience with the indicator.

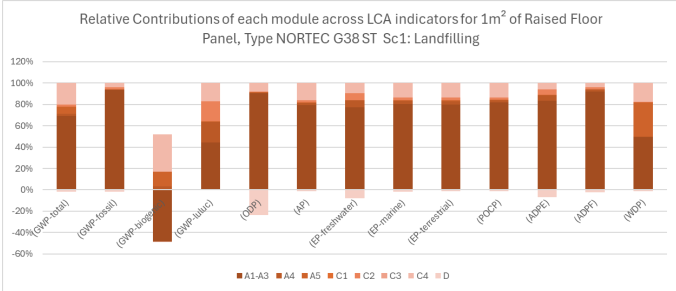
6. LCA: Interpretation

This Environmental Product Declaration (EPD) covers a specific product, Raised Floor panel, Type NORTEC G38 ST, with a declared unit of 53.72 kg/m². The assessment ensures that environmental performance is reported consistently for this product, taking into account its characteristics and typical use.
Total Global Warming Potential (GWP)
During the production phase (modules A1–A3), the GWP remains consistent across all scenarios. Additional contributions arise from transportation (module A4) and construction (module A5). The manufacturing processes from stages A1 to A5 remain the same, covering raw material extraction, production, and packaging disposal.
Scenario 0 – Reuse: End-of-life contributions are minimal, with emissions reduced by extending the product lifecycle.
Scenario 1 – Landfilling: At end-of-life, emissions increase due to the release of biogenic carbon during decomposition. Other stages remain stable, with minimal contributions from transportation and construction.
Note: In all scenarios, biogenic carbon is released at end-of-life in accordance with *EN 15804*.
Module D demonstrates the environmental benefits of recycling and reuse compared to landfilling. Recycling mitigates GWP by reintroducing materials into production, lowering raw material demand. Reuse provides the most significant GWP reduction by avoiding new product production and minimizing resource consumption.



GWP Fossil & GWP Biogenic
The contributions of fossil and biogenic carbon across all LCA indicators are presented for each scenario.
Scenario 0 – Reuse: Reuse strongly reduces both fossil and biogenic emissions. Production remains the main contributor to fossil emissions, while end-of-life stages contribute minimally to fossil emissions. Module D shows the greatest environmental benefits, highlighting the significant carbon savings achieved through extending the product's life.
Scenario 1 – Landfilling: Fossil emissions dominate, while biogenic carbon provides a negative contribution due to carbon sequestration. End-of-life stages show biogenic carbon release, while other stages remain stable. Module D provides limited benefits under landfilling conditions.





7. Requisite evidence

7.1 Formaldehyde
No formaldehyde is used in the manufacturing of this product.
7.2 MDI
No MDI adhesive systems are used in the manufacture of thisproduct. **7.3 Testing pretreatment of substances used**
The declared product does not contain "recycled wood" in its basic materials. Therefore, there is no basis for testingaccording to the Recycled Wood Regulation (AltholzVO).
7.4 Toxicity of Combustion Gases
The product is non-flammable, making the assessment of the toxicity of combustion gases irrelevant. A toxicological investigation has been conducted. **7.5 VOC Emissions**

AgBB overview of results (28 days [µg/m³])

Name	Value	Unit
TVOC (C6 - C16)	86	µg/m^3
Sum SVOC (C16 - C22)	<5	µg/m^3
R (dimensionless)	0.047	-
VOC without NIK	<5	µg/m^3
Carcinogenic Substances	<1	µg/m^3

AgBB overview of results (3 days [µg/m³])

Name	Value	Unit
TVOC (C6 - C16)	270	µg/m^3
Sum SVOC (C16 - C22)	<5	µg/m^3
R (dimensionless)	0.13	-
VOC without NIK	78	µg/m^3
Carcinogenic Substances	<1	µg/m^3

8. References

Standards
EN 15804
EN 15804:2012+A2:2019+AC:2021, Sustainability of construction works — Environmental Product Declarations — Core rules for the product category of construction products.

EN 12825
DIN EN 12825:2002-04, Raised Floors.

EN 13501-1
DIN EN 13501-1:2019-05, Classification of construction products and types into their fire behavior.

DIN EN 12825
DIN EN 12825:2002-04, Raised Floors, specifying the terminology, requirements, and test methods for raised access flooring systems used in construction.

EN ISO 14025
EN ISO 14025:2011, Environmental labels and declarations — Type III environmental declarations — Principles and procedures.

EN ISO 14044:2006
EN ISO 14044:2006, Environmental management — Life cycle assessment — Requirements and guidelines.

EN ISO 9001
EN ISO 9001, Quality management systems — Requirements.

EN ISO 50001
EN ISO 50001, Energy management systems — Requirements with guidance for use.
ISO 15686
ISO 15686, Buildings and constructed assets — Service life planning.

DIN SPEC 91484
Procedure to record building materials as a base to evaluate the potential for high-quality reutilization prior to demolition and renovation work.

Further References

European Waste Catalog (EWC)
European Waste Catalog (EWC), Commission Decision 2000/532/EC (as amended by Commission Decision 2014/955/EU).

LCA LCA for Experts
Sphera's LCA for Experts (GaBi) 10.7.1.28

LCA database
Sphera MLC (fka GaBi) CUP 2023.02

AgBB 2021
Procedure for the health-related evaluation of emissions of volatile organic compounds from construction products.

Biocidal Products Regulation

Regulation (EU) No 528/2012 of the European Parliament and of the Council of 22 May 2012 concerning the making available on the market and use of biocidal products, Official Journal of the European Union, 2012.

European Waste Catalog (EWC)
European Waste Catalog. Code 17 08 02 for materials based on gypsum not classified under 17 08 01.

REACH Regulation
European Parliament and Council. (2006). Regulation (EC) No1907/2006 concerning the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH). Official Journal of the European Union, L396, 1-849.

EN PCR part A
Calculation Rules for the Life Cycle Assessment and Requirements on the Project Report according to EN15804+A2:2019 /IBU PART A/. Version 1.4.

EN PCR part B
PCR Guidance-Texts for Building-Related Products and Services From the range of Environmental Product Declarations of Institute Construction and Environment eV(IBU)- Requirements on the PD for System floors, v10, Lastamended: 04/30/2024.

Federal Institute for Building, Urban Affairs and Regional Planning (BBSR): 353.211
Metal panelling: aluminium, steel, copper, zinc. Service lives of components. Service lives of components for Life Cycle Analyses in accordance with the evaluation system for sustainable building (BNB), in: German Federal Ministry for the Environment, Nature Conservation and Nuclear Safety (pub.), 2017

Publisher

Institut Bauen und Umwelt e.V.
Hegelplatz 1
10117 Berlin
Germany

+49 (0)30 3087748- 0
info@ibu-epd.com
www.ibu-epd.com

Programme holder

Institut Bauen und Umwelt e.V.
Hegelplatz 1
10117 Berlin
Germany

+49 (0)30 3087748- 0
info@ibu-epd.com
www.ibu-epd.com

Author of the life cycle assessment

Lindner Group
Bahnhofstraße 29
94424 Arnstorf
Germany

+49 (0) 8723 20 0
Nachhaltiges.Bauen@lindner-group.com
www.Lindner-Group.com

Owner of the Declaration

Lindner Group
Bahnhofstraße 29
94424 Arnstorf
Germany

+49 (0) 8723 20 0
Nachhaltiges.Bauen@lindner-group.com
www.Lindner-Group.com